

Programs based on arrays

1. Write a program to input the marks of 10 students in an array of integers and display the marks.
2. Write a program to count how many times a number is present in an array.
3. Write a program to subtract two arrays of the same size.
4. Write a program to input the sales made by a salesman in every month of a year and find out the total, average, maximum, and minimum sales.
5. Write a program to calculate the average of 10 values stored in an array and display all those values that are more than the calculated average.
6. Write a program that finds the locations and values of the largest and second largest element in a one-dimensional array.
7. Write a program to reverse an array of floats.
8. Write a program to input 20 values in an integer array and count the number of negative, positive, odd, and even values in the array.
9. Write a program to copy the last 5 elements of array b after the first 5 elements of array a. Assume the length of a and b is greater than 5.
10. Write a program to print all the prime numbers from an integer array and find out their sum.
11. Write a program to input 10 values in a float array and display all the values more than 75.
12. Write a program to delete an element from the array and shift the elements to the left.
13. Write a program to delete all the occurrences of a given value from the array and shift the remaining elements to the left of the array.
14. Write a program to delete duplicate elements from an array.
15. Write a program to insert a value in the array at the desired location.
16. Write a program to insert a value in a sorted array.
17. An array consists of 50 integers in the range of 1 to 25. Write a program that prints the number of times each integer occurs in the array.
18. Write a program to produce an array c containing all the elements of arrays a and b.
19. Write a program to delete all elements from an array between two specified positions.
20. Write a program to merge 4 arrays, a, b, c, and d in one array.

Two-dimensional arrays (matrix)

1. Write a program to input the values in a two-dimensional array of integers and display the values.
2. Write a program to find out the row sum and column sum of a two-dimensional array of integers.
3. Write a program to double all the elements of a matrix.

4. Write a program to print the left and right diagonal elements of a square matrix and find out their sum separately.
5. Write a program to find out the sum of non-diagonal elements of a square matrix.
6. Write a program to display the upper and lower triangle of a matrix.
7. Write a program to find out the sum and difference of two matrices.
8. Write a program to find out the product of two matrices.
9. Write a program to find out the sum of negative, positive, odd, and even integers separately from a two-dimensional array.
10. Write a program to display those elements of a two-dimensional array that are divisible by 10. Also find their sum.
11. Write a program to find out the sum of those numbers in a 2D array of integers that are divisible by 4 but less than 15.
12. Write a program to print all the prime numbers from a 2D array and find out their sum.
13. Write a program to transpose a square matrix.
14. Write a program to check the equality of two matrices.
15. Write a program to print all the elements of a square matrix except the diagonal elements.
16. Write a program to find the maximum and minimum of a matrix.
17. Write a program to print the squares of the diagonal elements of a matrix.
18. Write a program that finds the locations and values of the largest and second largest element in a two-dimensional array.
19. Write a program to check if a 2D array is a magic square.
20. Write a program to find out the trace of a square matrix.